BP-R Stage 3: interface continuation

7 October 2026 — static review; S3-G2-Q discussion; Stage 3 open

Status. The previous partial result is accepted. Reference A, upstream admission M.A and priority N.A remain closed. Neither compact route Q.A nor Q.B is adopted. These are analytical checks, not traffic simulation, parameter estimation, clearance validation or a gradient test.

1. Conservative count and event ownership

Let $F_{cb}^k \geq 0$ be actual continuous mass transferred from control volume c to b during time bin k, n_c^k its mass, and I_c^k, E_c^k its external arrival/exit mass. For road/ramp and entrance waiting volumes together,

$$n_c^{k+1} = n_c^k + I_c^k - E_c^k + \sum_b F_{bc}^k - \sum_b F_{cb}^k, \qquad M^{k+1} - M^k = I^k - E^k. \tag{1}$$

Internal transfers cancel. A descriptor adds no mass. The source's lateral transfer advances to the next longitudinal target; it is not advected twice. Receiving, sending and source step-size bounds remain separate requirements.

For stream s, $H_s(t)$ is actual cumulative transferred mass after the start, $m_* = 1$ vehicle-equivalent is the descriptor threshold, and $z_s \in [0, 1)$ is public initial threshold progress. At the stated source sampling instants, the number of new descriptors is

$$K_s(t) = \lfloor z_s + H_s(t)/m_* \rfloor, \qquad \sum_{s=1}^{S} K_s(t) \leq LP/m_* + S. \tag{2}$$

The bound is conditional on a finite acyclic directed mesh: every unit of the finite offered/initial population P crosses at most L descriptor-generating interfaces. S is the finite number of such streams. Initial descriptors are additional finite public state. The bound does not establish clearing or justify continuous event-time interpolation of sampled source births.

For a source-free lane segment with cumulative-count field $N(x, t)$, density $k = -N_x$, flow $q = N_t$, and constraint path $x_b(t)$, zero relative passing gives

$$\frac{d}{dt} N(x_b(t), t) = q(x_b, t) - \dot{x}_b \, k(x_b, t) = 0. \tag{3}$$

This is the local moving-boundary interface. Distributed lateral sources and junction events require count anchoring/reindexing; equation (3) is not a proof of their joined solution. Cell inventories, node counts and moving-path values must be consistent. No source coincidence perturbation is adopted.

2. A concrete dynamic compact source law

Let D_r, D_m be ramp and compatible mainline-group sending rates, R_0 the *ordinary, unmodified* target receiving rate, Q target FD capacity, $p \in (0,1)$ the approved fixed ramp priority, and $\delta \in [0,1)$ a fixed phenomenological capacity-drop fraction. All flows have units veh-equivalent/s. RM0010, Jin, Gan and Lebacque (2013), Section 6 equation (15), supplies

$$R_* = \min\{R_0,\ Q(1 - \delta\, \mathbf{1}_{\{D_r + D_m > R_0\}})\}. \tag{4}$$

Its priority merge transcription is

$$f_r = \min\{D_r, \max\{R_* - D_m, pR_*\}\}, \quad f_m = \min\{D_m, \max\{R_* - D_r, (1-p)R_*\}\}. \tag{5}$$

Use the already approved non-ramp subdivision. Densities evolve by equation (1), with flows formed from the model's own state; approach admission remains M.A. These wrapper/grouping adaptations are not the source's multilane theorem. The source law is phenomenological, not a derived disturbance kinetic law.

Local bounds follow from the previously qualified priority operator:

$$0 \le f_r \le D_r, \quad 0 \le f_m \le D_m, \quad f_r + f_m = \min\{D_r + D_m, R_*\} \le R_0. \tag{6}$$

Thus this is an independent dynamic conservative candidate, rather than only a mean-throughput formula. The trigger must use R_0, not already reduced R_*. Bounds do not prove predictive adequacy, joined reference validity or gradients.

Reviewer hand counterexample to the inactive match. Take no ramp entry or active moving constraint. Share the same current ordinary density/receiving state across models and assign

$$D_r = 0, \quad D_m = Q, \quad R_0 = 0.9Q, \quad \delta = 0.2: \qquad f_m^{\text{compact}} = 0.8Q < 0.9Q = f_m^{\text{unobstructed}}. \tag{7}$$

These values are an algebraic witness, not calibrated data or a simulated trajectory. The source model's own physical domain is not refuted. Its $\delta = 0$ limit gives the ordinary node; that different parameter limit does not prove equality when constraints stop binding at fixed physical parameters. Replacing the trigger to erase this discrepancy would be a new closure.

Q.A proposes qualifying a quasi-steady mean-service embedding while preserving the inactive criterion. Q.B would retain this native law and explicitly amend the compact admission criterion to a predictive-error diagnostic. Neither option is selected. A source-family prediction difference is not a pure state-removal theorem.

3. What can be proved about recovery and the TST tail

For the source acceleration on level terrain, suppose $a_0 > 0$ and $v_{\max} > u > 0$, where u is free speed. While realized speed $v \le u$,

$$a(v) = a_0(1 - v/v_{\max}) \ge a_{\min} = a_0(1 - u/v_{\max}) > 0, \qquad T_{\mathrm{rec}} \le (u - v_0)_+/a_{\min}. \tag{8}$$

The recovery bound applies *after downstream flow permits unrestricted acceleration*, with initial desired speed at least v_0. A discrete source update may add one time bin. Downstream congestion can postpone that start; finite descriptor count alone does not prove eventual release. These stated conditions are sufficient analytic conditions, not new adopted geometry.

After the last offered arrival, total inventory $M(t)$ is nonnegative and nonincreasing by conservation. Suppose the eventual joined law independently certifies constants $H > 0$ and $0 < \eta \le 1$ for all $t \ge T$:

$$M(t + H) \le (1 - \eta)M(t). \tag{9}$$

Then $M(T + jH) \le (1 - \eta)^j M(T)$. Bounding each interval by its start inventory and summing the geometric series yields

$$\int_T^\infty M(t)\,dt \le H \sum_{j=0}^{\infty} M(T + jH) \le \frac{HM(T)}{\eta}. \tag{10}$$

The right side has vehicle-equivalent seconds as units. It certifies a residual tail for complete TST without declaring a nonempty road cleared. The constants must be established from the joined law, not assumed from positive permission, finite population, fitted terminal values or the absence of newly arriving traffic.

Remaining obligations. The adapted reference's simultaneous-event coupling and whole draining certificate are not proved. The compact mean map still needs its own inputs, regimes and consistent coupled solution. Common rate-to-exit/TST leverage, policy dominance, state necessity and useful gradients remain unanswered. The current result is new interface/compact qualification evidence and a concrete S3-G2-Q discussion, not Stage 3 closure.

Source locations and status

Supplied Laval–Daganzo (2006), Sections 2.2 and Appendix A; Daganzo–Laval RR-2003-2, Sections 2–4; RM0010 (2013), Sections 4 and 6, `https://arxiv.org/pdf/1310.2660`. Node and moment wrappers here are declared review proposals. The tail inequality and inactive hand example are reviewer analysis. See the adjacent evidence and assessment Markdown records for source-access limits and the two candidate contracts.